

EXPERIMENT-4

DSA0606-DATA HANDLING AND VISUALIZATION

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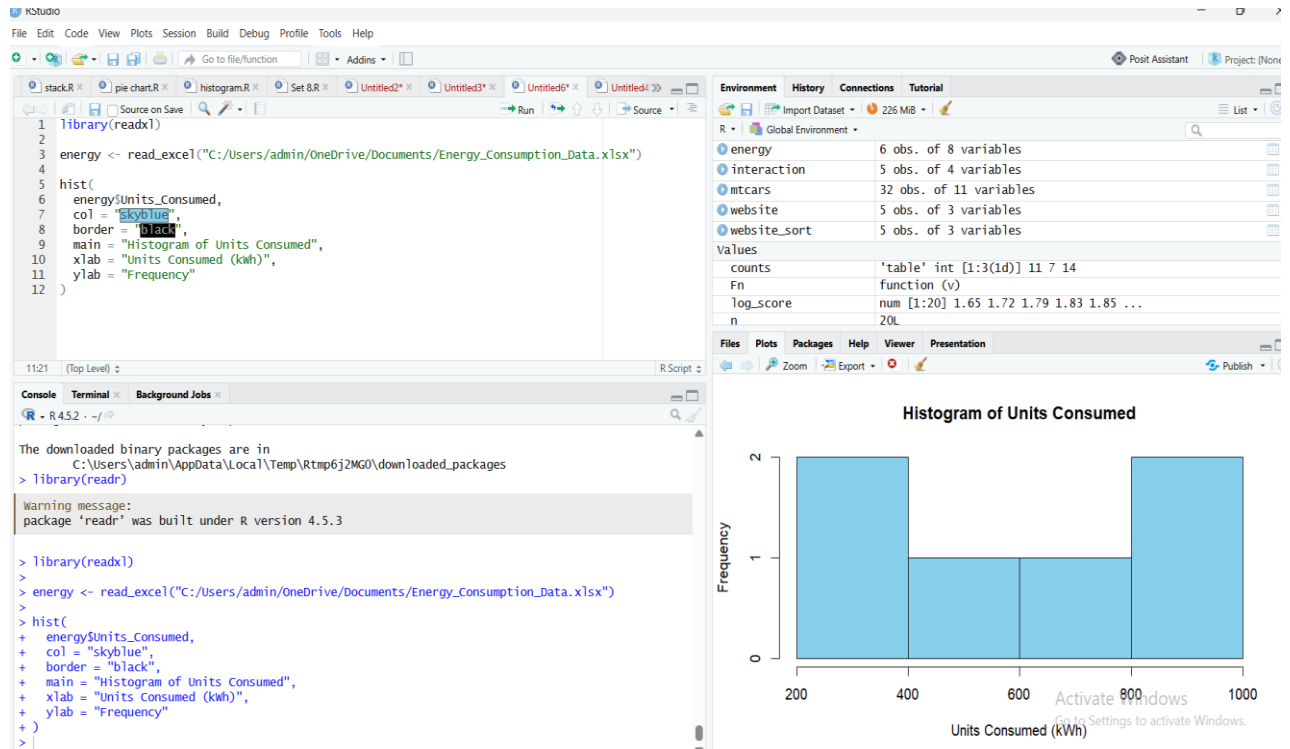
B.Tech (AI &DS)

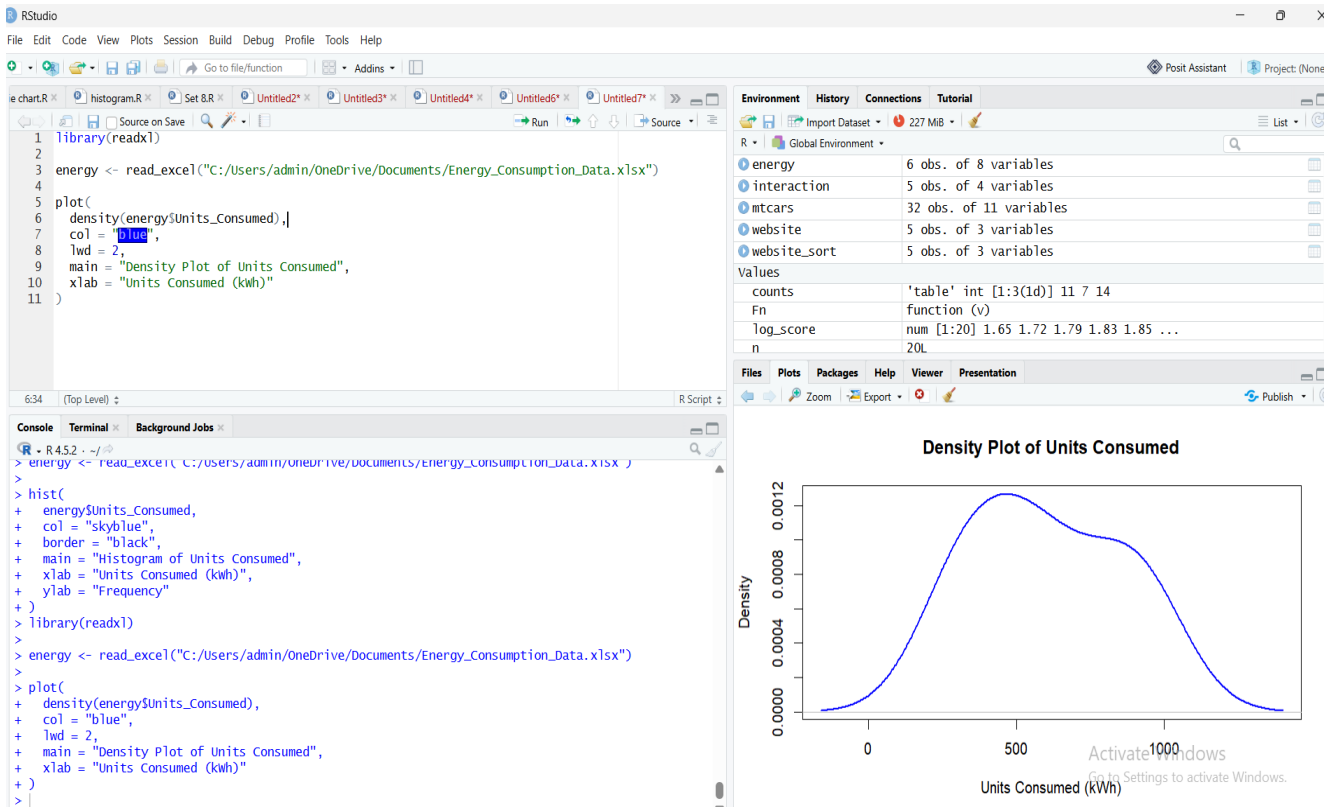
SET-14

Dataset: Energy_Consumption_Data.csv

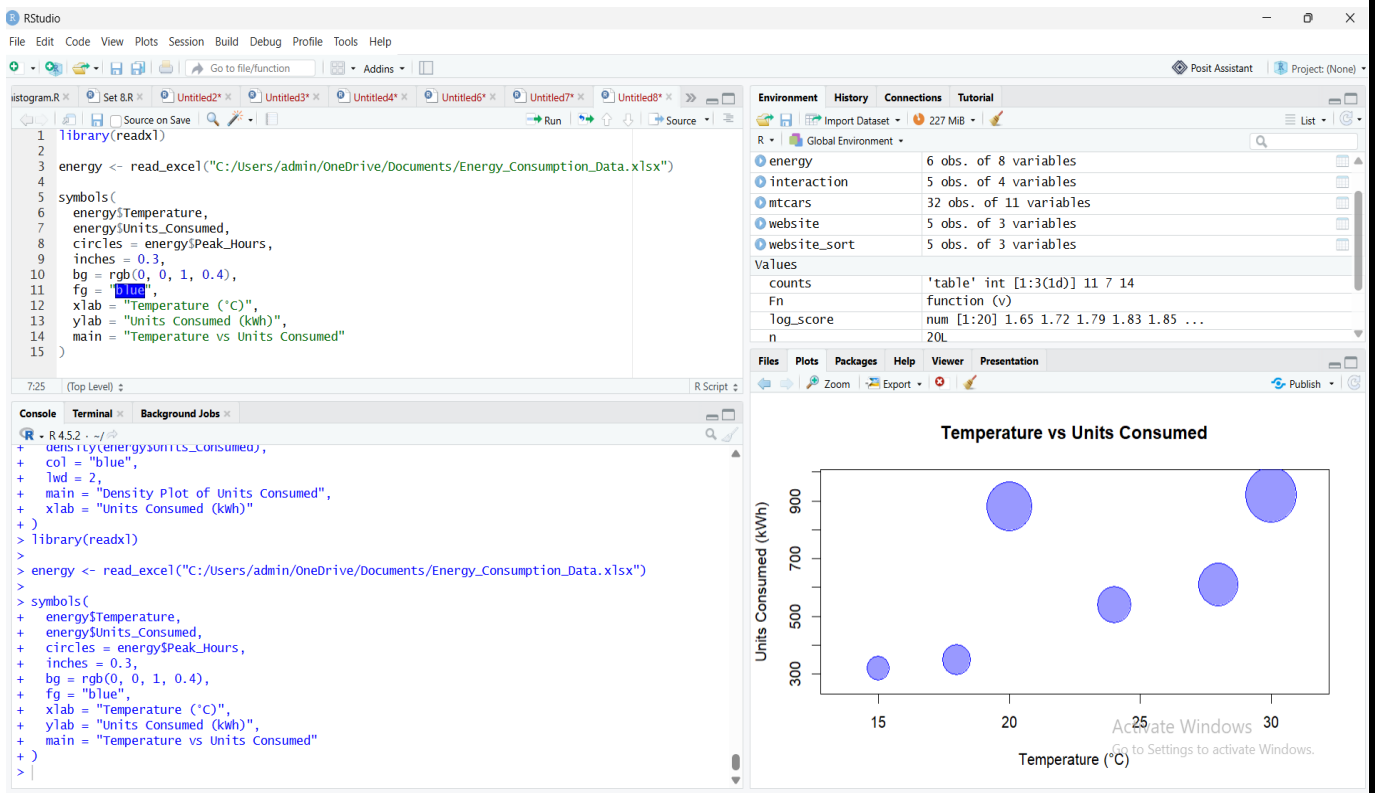
Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage (%)	Peak_Hour
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

1. Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

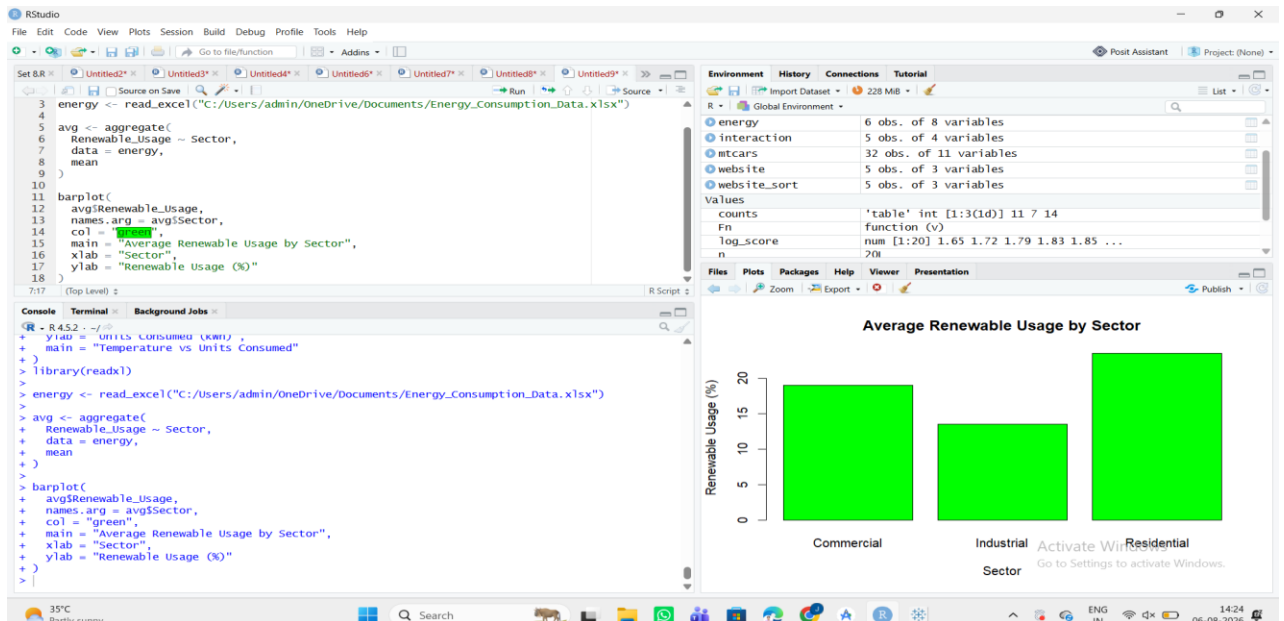




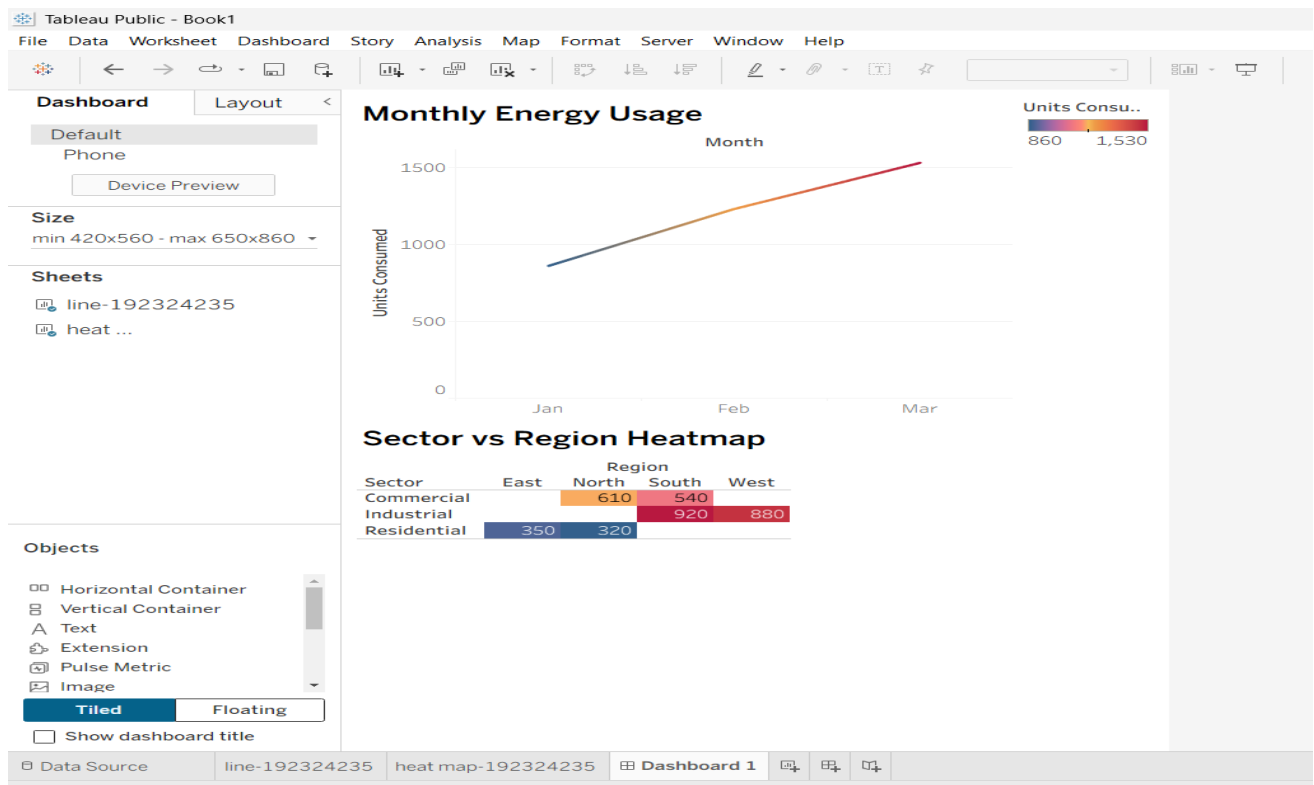
2. Using **R programming**, create a scatter plot between **Temperature** and **Units_Consumed**. Represent **Peak_Hours** using bubble size. Apply transparency.



3. Using **R**, calculate average **Renewable_Usage** for each **Sector**. Create a bar chart. Interpret renewable energy adoption.



4. Import **Energy_Consumption_Data.csv** into **Tableau**. Create a line chart for monthly energy usage trends. Create a heatmap (**Sector** vs **Region** vs **Units**). Add filters for **Month** and **Sector**. Design an interactive dashboard.



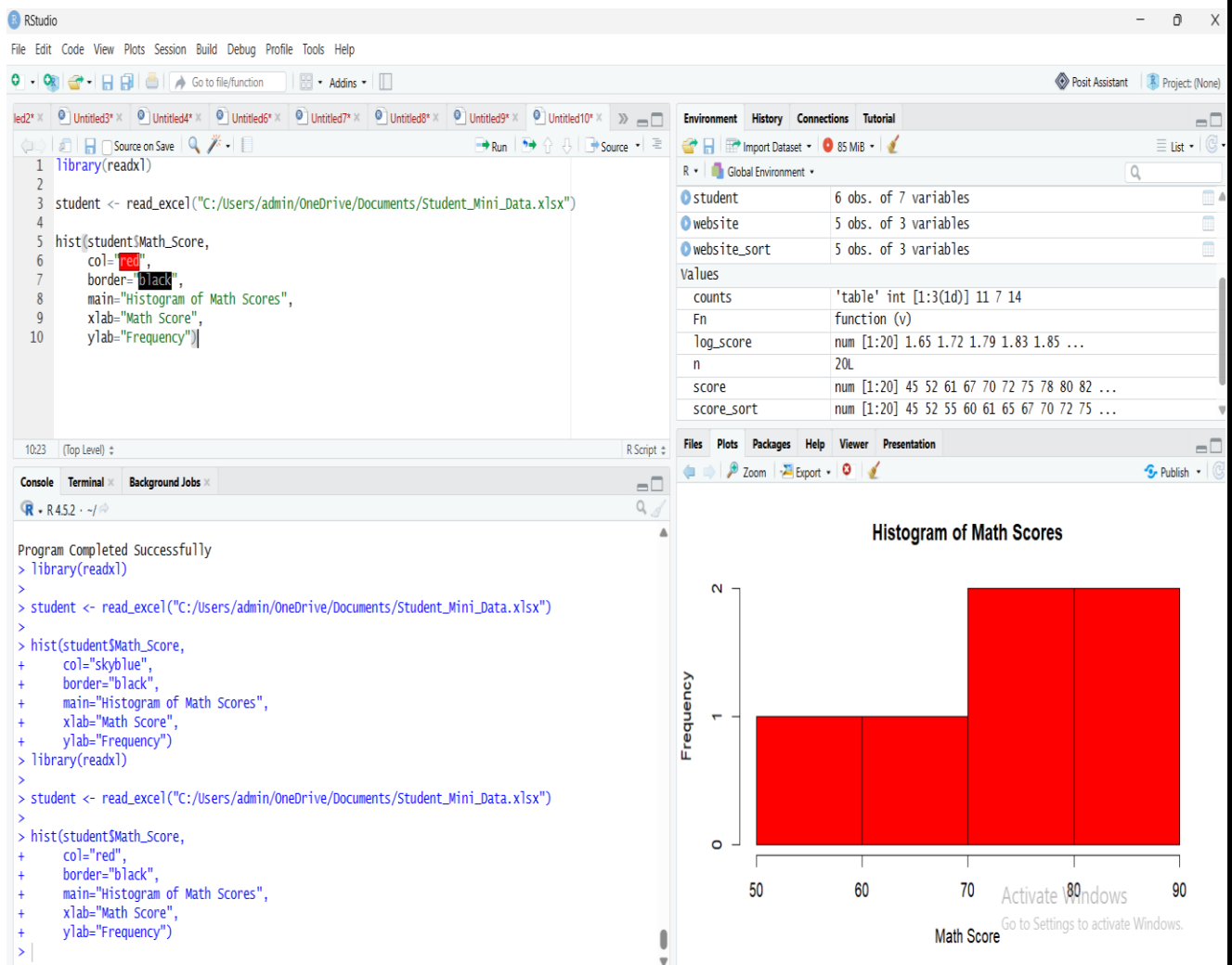
SET 15– Student Mini Data Analysis

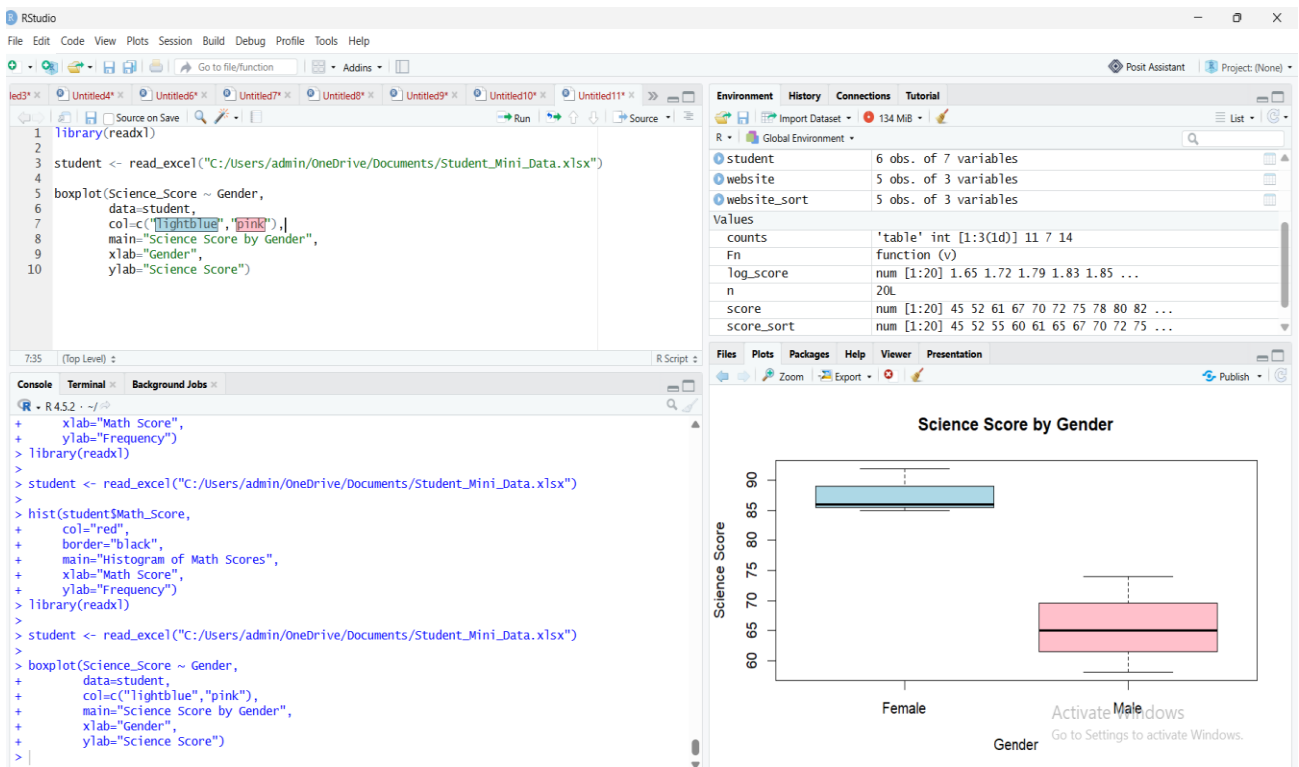
Dataset: Student_Mini_Data.csv

Student	Gender	Study_H	Attend	Math_S	Science_S	Exam_
S01	Ma	2.0	78	62	65	2025-0
S02	Fer	3.5	90	80	85	2025-0
S03	Ma	1.5	70	55	58	2025-02
S04	Fer	4.0	95	90	92	2025-02
S05	Ma	2.8	85	72	74	2025-03
S06	Fer	3.0	92	82	86	2025-03

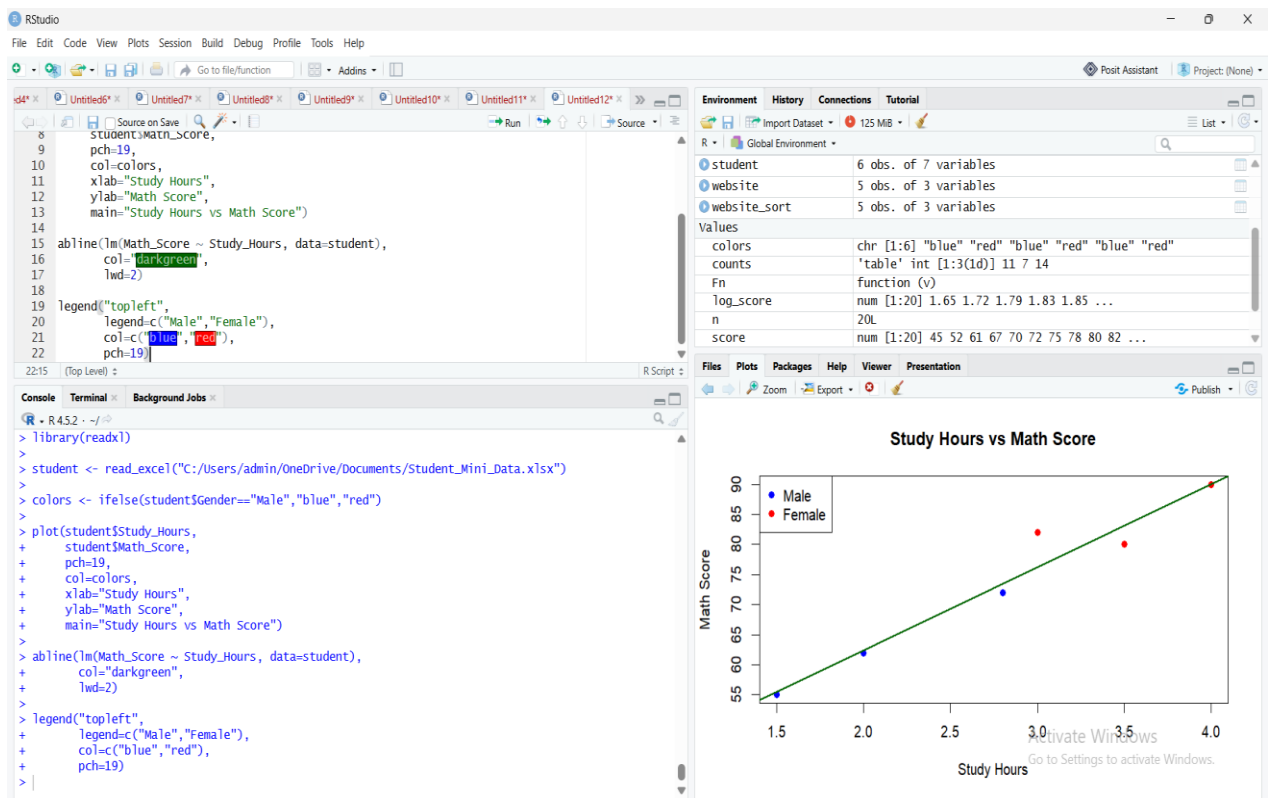
Questions

1. Import the dataset in R. Plot a histogram for **Math_Score**. Draw a boxplot of **Science_Score** by **Gender**. Add proper title and labels. Interpret the results.

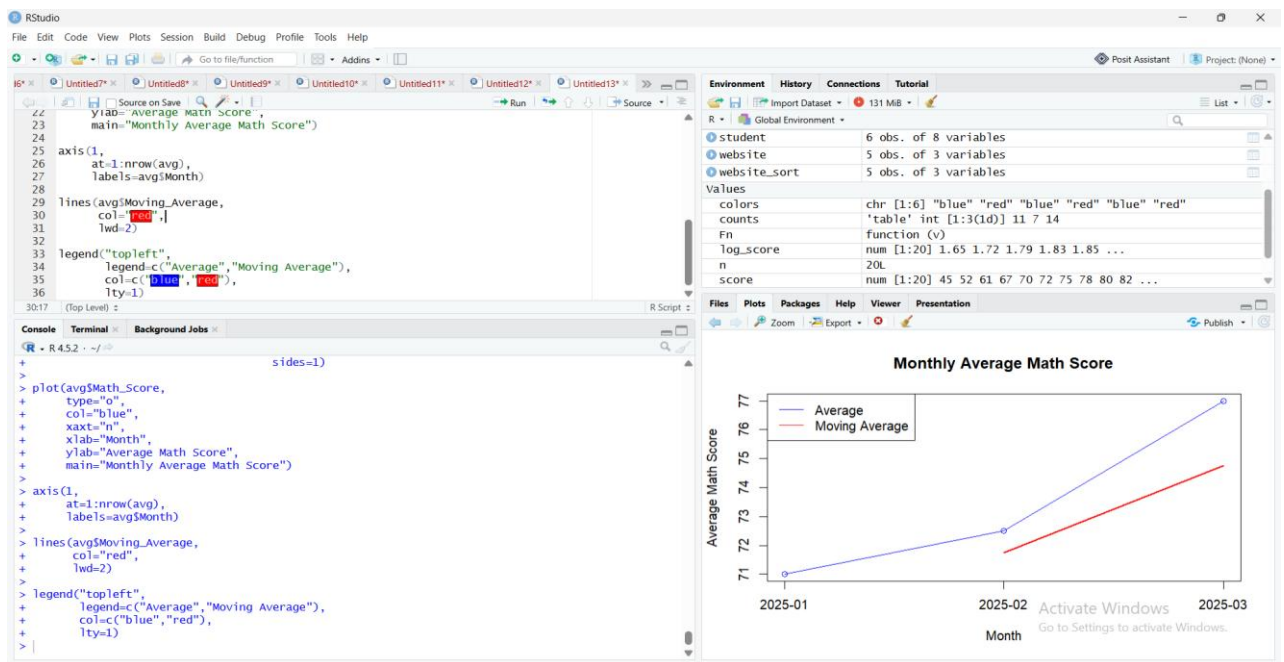




2. Create a scatter plot using R programming between **Study_Hours** and **Math_Score**. Use different colors for **Gender**. Add a regression line. Explain the relationship between study time and performance.



3. Using R, convert **Exam_Date** into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.



4. Import **Student_Mini_Data.csv** into Tableau. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for **Exam_Date**. Design a simple interactive dashboard.

